Assignment: Patient Management Dashboard (React + TypeScript)

Objective

Build a **Patient Management Dashboard** using **React with TypeScript** that allows users to add, view, search, and summarize patient records.

Problem Statement

A healthcare clinic wants a simple web application to manage patient records.

You need to build a **dashboard** where:

- New patients can be added
- Existing patients can be viewed in a grid
- Patients can be searched
- Total patient count is displayed

Functional Requirements

Add Patient

Create a form to capture:

- patientId (number)
- patientName (string)
- diagnosis (string)

Validations:

- Patient ID cannot be negative
- Patient name cannot be empty
- Show error messages if validation fails

View Patient List

Display all patients in a **table/grid**:

- Show:
    - "Still loading" if no data
    - "Patient Records" when data exists

Search Patients

Add a search box to filter patients:

- Search by:
  - Patient name
  - Diagnosis

Use **useMemo** for optimization

State Management

- Use **custom hooks**:
  - usePatients → manage patient list
  - usePatientCount → return total count

Technical Requirements

- Use **React + TypeScript**
- Functional components with hooks
- Use:
  - useState
  - useMemo

📁 **Suggested Folder Structure**

```
src/
 ├ components/
 │  ├ AddPatient.tsx
 │  ├ PatientGrid.tsx
 │  ├ PatientSummary.tsx
 ├ hooks/
 │  ├ usePatients.ts
 │  ├ usePatientCount.ts
 ├ types/
 │  ├ Patient.ts
 ├ App.tsx
```

Sample Data Model

```
export interface Patient {
  patientId: number;
  patientName: string;
  diagnosis: string;
}
```

Points to Remember

1) Use Vite for building React Application
2) Push the code in github
3) Provide access to ptmarytheresa@github.com
4) Share your git details over mail to learner.support@stackroute.in